

NET Centric Computing

Course Title: NET Centric Computing

Course no: BIT351

Nature of the Course: Theory + Lab

Semester: VI

Full Marks: 60+20+20

Pass Marks: 24 +8+8

Credit Hrs: 3

Course Description:

The course covers the concepts of cross-platform web application development using the ASP.NET Core MVC framework using C# programming Language.

Course Objectives:

The objective of this course is to understand the theoretical foundation as well as its practical aspects of ASP.NET Core web application framework and C# language features.

Course Contents:

Unit 1: Language Preliminaries (8Hrs.)

Overview of .Net framework, Compilation and execution of .Net applications, Constructor, Properties, Arrays and String, Indexers, Inheritance, use of “base” keyword, Method hiding and overriding, applying polymorphism in code extensibility, abstract class sealed class, interface, Delegate and Events, Partial class, Collections, Generics, File IO, Try statements and Exceptions, Attributes: Attribute Classes, Named and Positional Attribute Parameters, Attribute Targets, Specifying Multiple Attributes

Unit 2: Introduction to ASP.NET (3 Hrs.)

.NET and ASP.NET frameworks: .NET, .NET Core, Mono, ASP.NET Web Forms, ASP.NET MVC, ASP.NET Web API, ASP.NET Core, .NET Architecture and Design Principles, Compilation and Execution of .NET applications: CLI, MSIL and CLR, .NET Core in detail, .NET CLI: build, run, test and deploy .NET Core Applications

Unit 3: HTTP and ASP.NET Core (3 Hrs.)

HTTP, Request and Response Message Format, Common web application architectures, MVC Pattern, ASP.NET Core Architecture Overview, Projects, and Conventions, ASP.NET and ASP.NET MVC

Unit 4: Creating ASP.NET core MVC applications (7 Hrs.)

Setting up the Environment, Controllers and Actions: Create Controllers, Create Actions and Action Results Types, Rendering HTML with Views: Razor Syntax, Understanding Tag Helpers, Models: Binding and Validations, URL Routing and features, Web API Applications: API Controllers, JSON, Dependency Injection and IOC containers

Unit 5: Working with Database (6 Hrs.)

ADO.NET basics: Connection, Command, Reader and Adapter classes, Entity Framework (EF) Core, Object-Relational Mapper (ORM), Adding EF Core to an application: Choosing database provider, data models and data context, Querying and Saving data to database: Create, read, update and delete records

Unit 6: State Management on ASP.NET Core Application (4 Hrs.)

State Management on stateless HTTP, Server-side strategies: Session State, TempData, Using HttpContext, Cache Client-side strategies: Cookies, Query Strings, Hidden Fields

Unit 7: Client-side Development in ASP.NET Core (4 Hrs.)

Common client-side web technologies, JQuery, Forms and Validation, Single Page Application (SPA) Frameworks: Angular, React

Unit 8: Building Applications with React and ASP.NET Core (3 Hrs.)

Creating Forms in React, Making REST API Calls from React to ASP.NET Core for integration

Unit 9: Securing in ASP.NET Core Application (5 Hrs.)

Authentication: ASP.NET Core Identity, Adding authentication to apps and identity service configurations, Authorization: Roles, Claims and Policies, Securing Controllers and Action Methods, Common Vulnerabilities: Cross-site Scripting attacks, SQL Injection attacks, Cross-site Request Forgery (CSRF), Open Redirect Attacks

Unit 10: Hosting and Deploying ASP.NET Core Application (2 Hrs.)

App Servers and Hosting models: IIS, Nginx, Apache, ASP.NET Core Module, Kestrel, Docker and Containerization, Publish to Azure cloud

References:

1. C# 8.0 and .NET Core 3.0 – Modern Cross-Platform Development, Fourth Edition, by Mark J. Price, 2019
2. ASP.NET Core in Action, by Andrew Lock, 2018
3. Learning ASP.NET Core 2.0, Michel Bruchet, Jason De Oliveira, 2017
4. Learn ASP.NET Core 3 - Second Edition, Kenneth Yamikani Fukizi, Jason De Oliveira, Michel Bruchet, 2019
5. ASP.NET Core 5 and React By Carl Rippon

Laboratory works:

Students will have to complete a project covering most of the features of above course using ASP.NET Core and C#.

Homework Assignments:

Homework assignments can be given according to the course covered throughout the semester.

Computer Usage:

Windows PC or workstation installed with Latest Visual Studio Code or Visual Studio, .NET core SDK (3.0 or above) and Database: SQL Server.